GlobalMart Dimensional Model Design Document

1. Design Philosophy

The Silver layer provides clean data, but it is not inherently queryable for business users. Business users ask cross-functional questions—"Which vendors have the highest return rate?" or "Which region is most profitable?"—that require joining customers, products, and transactions simultaneously.

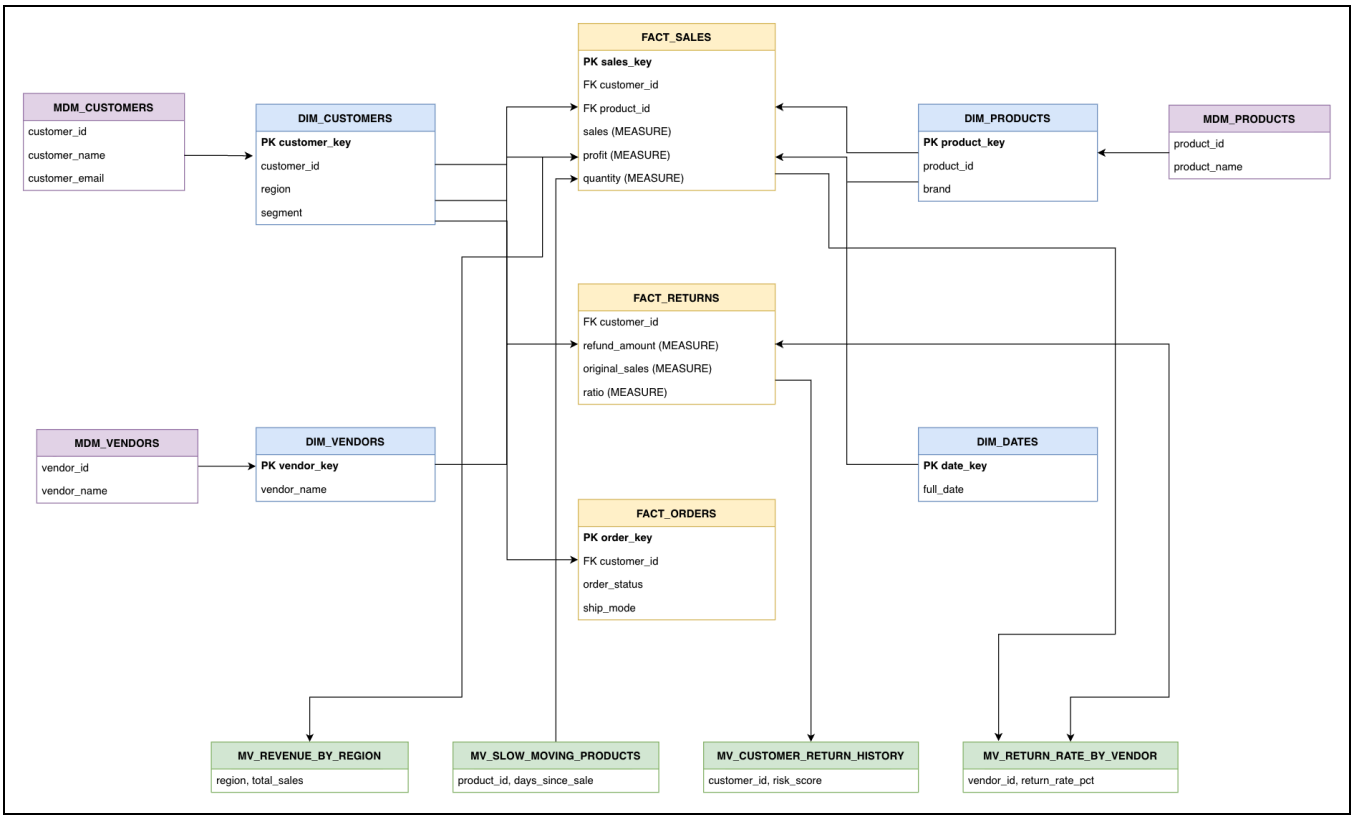
This model transitions GlobalMart from fragmented regional siloes into a **High-Performance Star Schema**. It is engineered to solve three specific business failures:

1. **Revenue Audit Failure:** Overcoming the ~9% revenue overstatement by unifying customer identities.
2. **Returns Fraud Exposure:** Mitigating the $2.3M annual loss by surfacing cross-regional return patterns.
3. **Inventory Blindspot:** Recouping 12–18% lost revenue through visibility into slow-moving stock.

Design Principles

- **Single-Event Grain:** Every row in a Fact table represents exactly one business event.
- **Flattened Headers:** Order header attributes are inherited by line items to maintain a pure Star Schema (avoiding Snowflake complexity).
- **SCD Type 1 MDM Integration:** Dimensions are sourced from MDM Golden Records to ensure a single version of the truth.

2. Dimensional Model Diagram (ERD)



3. Fact Table Specifications

3.1 fact_sales

- **Business Process:** The point-of-sale transaction at the line-item level.
- **Grain:** One row per unique order_id + product_id combination.
- **Measures:** sales, quantity, discount, profit.
- **Foreign Keys:** customer_key, product_key, vendor_key, order_date_key.

3.2 fact_returns

- **Business Process:** The processing of a customer refund.
- **Grain:** One row per return event per order_id.
- **Measures:** refund_amount, original_sales (approximated), return_to_sales_ratio.
- **Foreign Keys:** customer_key, product_key, vendor_key, return_date_key.

3.3 fact_orders

- **Business Process:** Accumulating snapshot of the order lifecycle.
- **Grain:** One row per order_id.
- **Measures/Attributes:** order_status, ship_mode, order_purchase_timestamp, order_delivered_customer_timestamp.
- **Foreign Keys:** customer_id, vendor_id.

4. Dimension Table Specifications

Dimension	Primary Key	Source Silver Table	Key Attributes
dim_customers	customer_key	mdm.customers (via silver.customers)	customer_id, customer_name, region, segment, email
dim_products	product_key	mdm.products (via silver.products)	product_id, product_name, brand, categories
dim_vendors	vendor_key	mdm.vendors (via silver.vendors)	vendor_id, vendor_name
dim_dates	date_key	Generated Calendar (Ref: silver.orders)	full_date, year, quarter, month_name, is_weekend

5. MDM Approach & Entity MDM

The transition from Silver to Gold is anchored by an **MDM (Master Data Management) Layer**. Unlike standard deduplication which simply picks the "latest" record, our MDM engine performs **Column-Level Survivorship** to build a high-fidelity "Golden Record."

The "Best of All Worlds" Approach

We implemented a multi-tiered survivorship strategy to resolve regional conflicts:

- **Source Priority (Identity Preservation):** For names and regions, we prioritized "Region 1" and "Region 2" as the systems of record. This prevents the overlapping customer IDs that caused the **9% revenue overstatement**.
- **Most Recent (Dynamic Attributes):** For addresses and phone numbers, we utilized a chronological _load_timestamp check. If a customer updated their address in a "Region 4" file, it correctly overrides the older "Region 1" data.
- **Most Complete (Completeness Maximization):** For contact data like emails, the engine identifies the most complete (non-null) value across all 6 regional sources. This ensures a customer profile in Gold is more detailed than any single source system.

By feeding these Golden Records into dim_customers, our Star Schema provides auditors with a single, reconcilable registry that finally bridges the gap between disconnected regional fragments.

6. Addressing Business Failures

GlobalMart Failure	Serving Layer MV	Design Solution
Revenue Audit	mv_revenue_by_region	**Unified Customer Identity:** The MDM-backed dim_customers eliminates regional ID overlaps. Grouping fact_sales by the unified region attribute provides a reconcilable registry for auditors.
Returns Fraud	mv_customer_return_history	**Cross-Regional History:** fact_returns joins to the unified dim_customers. This MV aggregates return volume across all 6 regions, flagging high-risk patterns instantly via fraud_risk_score.
Returns Fraud	mv_return_rate_by_vendor	**Vendor Quality Check:** Joins sales and returns facts to vendors. Identifies suppliers with high return volumes, allowing the business to renegotiate contracts or audit product quality.
Inventory Blindspot	mv_slow_moving_products	**Vendor/Product Velocity:** By connecting fact_sales to dim_products and dim_vendors, this MV highlights items with zero sales velocity, enabling timely regional redistribution.